

Proofs and Problem Solving

Week 7: Congruence

Notes based on Martin Liebeck's

A Concise Introduction to Pure Mathematics

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Week 7: Modular Arithmetic

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12. Modular Arithmetic

Is $30^{99} + 61^{100}$ divisible by 31? This seems like a daunting question. However, in a couple of pages, you'll learn a trick that will make this and other problems involving divisibility of very large numbers much easier. This method involves *modular arithmetic*.

The framework of modular arithmetic as we are familiar with it was first developed by Gauss in his book *Disquisitiones Arithmeticae* in 1801, whereas many fundamental results that are now stated in terms of congruences were in fact proven much earlier; for example, Fermat's Little Theorem below was proven in 1640, though now it is standard to formulate it using congruences.

Despite its classical roots, modular arithmetic is important for both mathematics and applications. In mathematics, it is fundamental for studying *abstract algebra*, which you will have opportunities to learn more about in future courses. Modular arithmetic is also a cornerstone of modern cryptography (Chapter 15 in Liebeck's book has a good introduction to *public key cryptography*).

12.1 Arithmetic modulo m

Definition 12.1 For $m \in \mathbb{N}$ and $a, b \in \mathbb{Z}$, we write $a \equiv b \pmod{m}$ or sometimes $a \equiv b \pmod{m}$ (read a is congruent to b modulo m) if there is some $q \in \mathbb{Z}$ such that $b - a = qm$.

The notion of congruence can be expressed in several different ways, as given in the following theorem.

Theorem 12.1 Let $m \in \mathbb{N}$ and $a, b \in \mathbb{Z}$

The following are equivalent:

- $a \equiv b \pmod{m}$;
- m divides $a - b$;
- a and b have same remainder when divided by m ;
- $b = qm + a$ for some $q \in \mathbb{Z}$.

Example 12.1 $32 = 2 \pmod{10}$. $82 = 1 \pmod{3}$.

Theorem 12.2 We have the following properties for integers $a, b, c \in \mathbb{Z}$ and $m \in \mathbb{N}$:

- (a) $a = a \pmod{m}$.
- (b) If $a = b \pmod{m}$, then $b = a \pmod{m}$.
- (c) If $a = b \pmod{m}$ and $b = c \pmod{m}$, then $a = c \pmod{m}$.

Proof. (a) Since m divides $a - a = 0$, we have that $a = a \pmod{m}$.

(b) If $a = b \pmod{m}$, then m divides $(a - b)$, so m divides $(b - a)$ as well, hence $b = a \pmod{m}$.

(c) Suppose $a = b \pmod{m}$ and $b = c \pmod{m}$. Then m divides $a - b$, so $a - b = km$ for some integer k . Moreover, m divides $b - c$, so $b - c = jm$ for some integer j . Thus,

$$a - c = a - b + b - c = km + jm = (k + j)m,$$

hence m divides $(a - c)$, thus $a = c \pmod{m}$. ■

Theorem 12.3 Suppose $a = x \pmod{m}$ and $b = y \pmod{m}$. Then

- (a) $-a = -x \pmod{m}$;
- (b) $a + b = x + y \pmod{m}$;
- (c) $ab = xy \pmod{m}$;
- (d) for any natural number k , one has $a^k = x^k \pmod{m}$.

Proof. We have that m divides $a - x$ and m divides $b - y$

(a) It is clear that m divides $-(a - x) = (-a) - (-x)$, so that $-a = -x \pmod{m}$.

(b) The Linear Combination Lemma implies that m divides

$$(a - x) + (b - y) = (a + b) - (x + y),$$

and so $a + b = x + y \pmod{m}$.

(c) $a = x \pmod{m}$ implies $a = x + pm$ for some $p \in \mathbb{Z}$ and $b = y \pmod{m}$ implies $b = y + qm$ for some $q \in \mathbb{Z}$. Thus,

$$ab = (x + pm)(y + qm) = xy + pmy + xqm + pqm^2 = xy + m(py + xq + pqm),$$

and so $ab = xy \pmod{m}$ by Theorem 12.1(3).

(d) This can be proven by induction. For the base case of $k = 1$, this is immediate. Now suppose $a^k = x^k \pmod{m}$ for some $k \geq 1$. Then by part (c) of this theorem with $b = a^k$ and $y = x^k$ and using the fact that $a = x \pmod{m}$, we have

$$a \cdot a^k = x \cdot x^k \pmod{m}.$$

Thus,

$$a^{k+1} = x^{k+1} \pmod{m}.$$

This proves the induction and hence the theorem. ■

This result proves that many of the algebraic manipulations we enjoy performing on integers – addition, subtraction, and multiplication – can be done modulo m systematically. This simplifies many challenging-looking problems. Let's revisit our earlier example: is

$30^{99} + 61^{100}$ divisible by 31? Note that $30 = 31 - 1 = -1 \pmod{31}$ and $61 = 2 \cdot 31 - 1 = -1 \pmod{31}$, and so

$$30^{99} + 61^{100} = (-1)^{99} + (-1)^{100} = -1 + 1 = 0 \pmod{31}.$$

Indeed, 31 divides $30^{99} + 61^{100}$.

Theorem 12.3 says we can add and multiply modulo m . Can we also divide? That is, if $ad = bd \pmod{m}$, do we also have $a = b \pmod{m}$? This is *not* always the case. Perhaps the simplest example is the following: $2 = 0 \pmod{2}$, but we cannot divide this equation by 2, because $1 \not\equiv 0 \pmod{2}$. More subtly, $12 = 24 \pmod{6}$, but here we cannot divide by 3, because $4 \not\equiv 8 \pmod{6}$. The following theorem says when we can divide in this way.

Proposition 12.4 Let a and m be coprime. If $x, y \in \mathbb{Z}$ and $xa = ya \pmod{m}$, then $x = y \pmod{m}$. In particular, if p is prime and p does not divide a , then $xa = ya \pmod{p}$ implies that $x = y \pmod{p}$.

Proof. Suppose a and m are coprime and $xa = ya \pmod{m}$. Then m divides $xa - ya = (x - y)a$. Since m and a are coprime, m divides $x - y$, and so $x = y \pmod{m}$ by definition. The second part of the proposition follows from the first. ■

Example 12.2 Find an integer $x \in \{0, 1, \dots, 6\}$ so that $4^6 = x \pmod{7}$.

Let's look at some powers of 4 and see what they are modulo 7.

$$4^2 = 16 = 14 + 2 = 2 \pmod{7}$$

$$4^4 = (4^2)^2 = 2^2 = 4 \pmod{7}.$$

Thus,

$$4^6 = 4^2 \cdot 4^4 = 2 \cdot 4 = 8 = 1 \pmod{7}.$$

This is certainly much quicker than computing 4^6 by hand and then doing long division!

12.2 Solving linear equations modulo m

When can we solve $ax = b \pmod{m}$ for x ? Before we even start, it's good to have a test to see if there is actually a solution:

Theorem 12.5 The equation $ax = b \pmod{m}$ has a solution if and only if $\gcd(a, m)$ divides b .

Proof. An integer x is a solution to $ax = b \pmod{m}$ if and only if $ax = b - my$ for some integer y . In other words, $ax = b \pmod{m}$ has a solution if and only if b can be written as $ax + my$ for some integers x and y .

If $b = ax + my$, then the Linear Combination Lemma ensures that $\gcd(a, m)$ divides b .

Conversely, assume that $\gcd(a, m)$ divides b , so that $b = h\gcd(a, m)$ for some integer h . Bezout's identity implies that there exist integers x' and y' such that $\gcd(a, m) = ax' + my'$. Hence if $x = hx'$ and $y = hy'$, then

$$b = h\gcd(a, m) = ax + my,$$

as desired. ■

Example 12.3 Find $x \in \{0, 1, \dots, 13\}$ so that $18x = 10 \pmod{14}$.

We can see that $\gcd(18, 14) = 2$ and 2 divides 10, so there is at least one solution. Now, we can first use the Euclidean Algorithm to express 2 as an integer linear combination of 14 and 18: since $18 - 14 = 4$ and $14 - 3 \cdot 4 = 2$ is the gcd, we can write

$$2 = 14 - 3 \cdot 4 = 4 \cdot 14 - 3 \cdot 18.$$

Since the factor relating 10 to 2 is 5, we can multiply the above equation by 5 to find

$$10 = 5 \cdot 2 = 20 \cdot 14 - 15 \cdot 18.$$

Thus, -15 is a solution. To check, let $x = -15$, so $18 \cdot (-15) = -270 = 10 + (-280) = 10 + (-20) \cdot 14$, and $18x = 10 \pmod{14}$.

There are several ways of making this argument simpler. First, we can observe that $4 = 18 \pmod{14}$. Thus, $18x = 4x$, so $18x = 10$ if and only if $4x = 10$. The gcd is still $\gcd(4, 14) = 2$ which still divides 10, so there is a solution. Applying the Euclidean Algorithm, $2 = 14 - 3 \cdot 4$. Multiplying by 5,

$$10 = 5 \cdot 14 - 15 \cdot 4.$$

Thus, $a = -15$ is a solution of $10 = 4a \pmod{14}$. Since $-15 = 13 \pmod{14}$, it is also true that $x = 13$ is a solution of $4x = 10 \pmod{14}$. To check, observe $18 \cdot 13 = 234 = 10 + 224 = 10 + (16) \cdot 14$, so $x = 13$ is a solution of $4x = 10$.

Note that when finding a solution to $a = bx \pmod{m}$, any integer x is congruent modulo m to a number in $\{0, 1, \dots, m-1\}$. Thus, whenever we ask for a solution, we prefer to write it as one of these integers.

■ **Exercise 12.1** Find x so that $7x = 8 \pmod{12}$.

12.3 $\mathbb{Z}/m$

In this section we introduce a set $\mathbb{Z}/m$ that will be important in your future algebra classes. Just like the integers, the rationals, the reals, and the complex numbers, it is what is called a *ring*: it is equipped with addition, subtraction, and multiplication operations. Unlike $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$, the set $\mathbb{Z}/m$ is *finite*.

As a set,

$$\mathbb{Z}/m = \{0, 1, \dots, m-1\}.$$

On this set we shall define addition and multiplication operations: for any $x, y \in \mathbb{Z}/m$,

$$x + y = z \text{ where } z \in \{0, 1, \dots, m-1\} \text{ is the unique element such that } x + y = z \pmod{m},$$

and

$$x \cdot y = z \text{ where } z \in \{0, 1, \dots, m-1\} \text{ is the unique element such that } x \cdot y = z \pmod{m},$$

There is an additive inverse:

$$-x = m - x$$

(Some books write $\mathbb{Z}_m$ for $\mathbb{Z}/m$, but most algebraists reserve this notation for a very different object!!)

Example 12.4 Let's write the addition and multiplication tables for $\mathbb{Z}/5 = \{0, 1, 2, 3, 4\}$.

+	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

$\times$	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

We can check that for $m \in \mathbb{N}$, $\mathbb{Z}/m$ has all the nice algebraic properties of being a field (recall this definition from Week 1) apart from perhaps property (M4), which said that each nonzero element must have a multiplicative inverse. An element $x \in \mathbb{Z}/m$ is *invertible* if there exists $y \in \mathbb{Z}/m$ so that $x \cdot y = 1$.

If m is composite, then not every nonzero element $\mathbb{Z}/m$ is invertible. Consider the multiplication table of $\mathbb{Z}/4$:

$\times$	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	0	2
3	0	3	2	1

As you can see, $2x = 1$ has no solution modulo 4.

Observe that an element $a \in \mathbb{Z}/m$ is invertible if and only if the equation $ax = 1 \pmod{m}$ admits a solution. Such a solution exists if and only if $\gcd(a, m)$ divides 1, hence if and only if a and m are coprime. Thus the invertible elements of $\mathbb{Z}/m$ are precisely those elements that are coprime to m . In particular, if p is prime, then every nonzero element of $\mathbb{Z}/p$ is coprime to p , so we deduce the following.

Proposition 12.6 Let $m \geq 2$ be a natural number. Then $\mathbb{Z}/m$ is a field if and only if m is prime.

■ **Exercise 12.2** How many invertible elements are there in $\mathbb{Z}/81$?

Just like there is a notion of inverses for $\mathbb{Z}/m$, we also have a notion of a root. We say $y \in \mathbb{Z}/m$ is the n -th root of $x \in \mathbb{Z}$ if $x = y^n$.

Example 12.5 In $\mathbb{Z}/5$, does 2 have a cube root? Let us test the values:

- $0^3 = 0$.
- $1^3 = 1$.
- $2^3 = 3$ since $2^3 = 8 = 3 \pmod{5}$.
- $3^3 = 2$ since $3^3 = 27 = 2 \pmod{5}$.

- $4^3 = 4$ since $4^3 = (-1)^3 = -1 = 4 \pmod{5}$.
Thus, the unique cube root of 2 is 3.

However, not all numbers in every $\mathbb{Z}/m$ have roots! Confirm for yourself that $2 \in \mathbb{Z}/4$ has no cube root.

12.4 Fermat's Little Theorem

The great thing about congruence is that arithmetic becomes easier since you can replace big numbers with smaller numbers that are congruent. The following is a little but powerful theorem due to Fermat that allows you to eliminate large powers if you are working modulo some prime.

Theorem 12.7 — Fermat's Little Theorem (FLT). Let a be an integer, and let p be a prime number. If p does not divide a , then

$$a^{p-1} = 1 \pmod{p}.$$

Proof. Without loss of generality, we may assume that $a \in \{1, \dots, p-1\}$. We consider the numbers $a, 2a, 3a, \dots, (p-1)a$. We first observe that each of these numbers is distinct modulo p ; that is, for each $c, d \in \mathbb{Z}/p$, one has $ca = da \pmod{p}$ if and only if $c = d$. This follows from Proposition 12.4.

If $\bar{a}, \bar{2a}, \dots, \overline{(p-1)a}$ are the remainders of $a, 2a, \dots, (p-1)a$ after dividing by p , then we deduce that the sets $\{\bar{a}, \bar{2a}, \dots, \overline{(p-1)a}\}$ and $\{1, 2, \dots, p-1\}$ are equal. Consequently,

$$a \cdot 2a \cdots (p-1)a = \bar{a} \cdot \bar{2a} \cdots \overline{(p-1)a} = 1 \cdot 2 \cdots (p-1) \pmod{p}.$$

Thus

$$(p-1)! a^{p-1} = (p-1)! \pmod{p}.$$

Now since p does not divide $(p-1)!$, Proposition 12.4 implies that $a^{p-1} = 1 \pmod{p}$, as desired. ■

Example 12.6 Find $5^{100} \pmod{7}$.

Since 7 does not divide 5, the FLT says $5^6 = 1 \pmod{7}$. Thus, we can shave off any power of 5 from 5^{100} that is a multiple of 6. Thus,

$$5^{100} = 5^{96} 5^4 = (5^6)^{16} 5^4 = 1^{16} 5^4 = 5^4 \pmod{7}.$$

Then the rest is as before:

$$5^2 = 25 = 4 \pmod{7}$$

$$5^4 = (5^2)^2 = 4^2 = 16 = 2 \pmod{7}.$$

Fermat's Little Theorem gives us a way of solving a large class of polynomial equations mod m :

Theorem 12.8 Let $n \in \mathbb{N}$ and p be a prime number. Assume n and $p-1$ are coprime and p does not divide b . Then the equation

$$x^n = b \pmod{p}. \tag{12.4.1}$$

has exactly one solution $x \in \{1, \dots, p-1\}$.

Proof. Since n and $p-1$ are coprime, there are integers s, t so that $sn + t(p-1) = 1$.

Let us prove the uniqueness. If $x_1, x_2 \in \{1, \dots, p-1\}$ are two such solutions, then $x_1^n = x_2^n \pmod{p}$, and so for $i \in \{1, 2\}$,

$$x_i = x_i^{sn+t(p-1)} = x_i^{sn} x_i^{t(p-1)} = b^s (x_i^t)^{p-1} \pmod{p}.$$

Since p does not divide x_i^t , Fermat's Little Theorem tells us that each $x_i = b^s \pmod{p}$, whence $x_1 = x_2$.

Now let us prove the existence. Let x be the remainder after dividing b^s by p . Now in $\mathbb{Z}/p$,

$$x^n = (b^s)^n = b^{ns} = b^{1-t(p-1)} = b \cdot (b^{p-1})^{-t}.$$

(Note that this makes sense, since p does not divide b , so b is invertible.) Since p does not divide b , Fermat's Little Theorem implies that $x^n = b \pmod{p}$. ■

Notice that even though the choice of s, t is not unique, the remainder after dividing b^s by p is unique. In other words, if s_1, t_1, s_2, t_2 are integers such that for $i \in \{1, 2\}$, we have

$$s_i n + t_i (p-1) = 1,$$

then $b^{s_1} = b^{s_2} \pmod{p}$. To see this directly, note that in $\mathbb{Z}/p$, we have by FLT:

$$b^{s_1} = (b^{s_1})^{s_2 n + t_2 (p-1)} = b^{s_1 s_2 n} (b^{s_1 t_2})^{p-1} = b^{s_1 s_2 n},$$

and by the same reasoning:

$$b^{s_2} = (b^{s_2})^{s_1 n + t_1 (p-1)} = b^{s_1 s_2 n} (b^{s_2 t_1})^{p-1} = b^{s_1 s_2 n}.$$

We now have a blueprint to how to solve equations of the form (12.4.1) when n and $p-1$ are coprime and p does not divide b .

Example 12.7 Find $x \in \{0, 1, \dots, 16\}$ so that $x^7 = 4 \pmod{17}$.

Since $\gcd(7, 17-1) = \gcd(7, 16) = 1$, we can use the Euclidean Algorithm to show that $7 \cdot 7 = 1 + 3 \cdot 16$. Thus, if x solves the above equation, then $4 = x^7$, and so

$$4^7 = (x^7)^7 = x^{7 \cdot 7} = x^{1+3 \cdot 16} = x \cdot (x^{16})^3 = x \cdot (1)^3 = x.$$

Thus, $x = 4^7$, so we need to find a representative in $\{0, 1, \dots, 16\}$. Note that $4^2 = 16 = -1 \pmod{17}$, so

$$4^7 = (4^2)^3 \cdot 4 = (-1)^3 \cdot 4 = -4 = 13 \pmod{17}.$$

Thus, $x = 13$ is the unique solution in $\{0, 1, \dots, 16\}$.

Not all polynomial equations can be solved using the above method:

Example 12.8 Find a solution to $x^{22} = 3 \pmod{11}$.

Note that $\gcd(22, 11-1) = \gcd(22, 10) = 2$, so we can't use the method in Theorem 12.8. However, let's suppose x is a solution to the above equation and try to narrow down what it must be. Note that 11 does not divide x (otherwise $x = 0 \pmod{11}$ and $x^{22} = 0 \pmod{11}$). Thus, FLT implies $x^{10} = 1 \pmod{11}$. Hence,

$$x^{22} = x^2 (x^{10})^2 = x^2.$$

Now we just need to solve $x^2 = 3 \pmod{11}$. Again, $\gcd(2, 11-1) = \gcd(2, 10) = 2$, so we still can't use the method of Theorem 12.8. But now the power 2 is small enough we can just try values $x \in \{0, 1, \dots, 10\}$ and see what works:

- $1^2 = 1$
- $2^2 = 4$
- $3^2 = 9$
- $4^2 = 16 = 5 \pmod{11}$.
- $5^2 = 25 = 3 \pmod{11}$.

Thus, $x = 5$ is a solution to $x^2 = 3 \pmod{11}$:

$$5^{22} = 5^2(5^{10})^2 = 3 \cdot (1)^2 = 3.$$

Note that if we can't use Theorem 12.8, then the solution may not be unique. In this example, note that $x = 6$ is also a solution

Example 12.9 Find $x \in \{0, 1, \dots, 10\}$ that solves $x^3 = 9 \pmod{11}$.

Note $\gcd(3, 11-1) = \gcd(3, 10) = 1$, so this tells us that the method of Theorem 12.8 should work in finding the unique solution. Note also that $7 \cdot 3 = 21 = 1 + 2 \cdot 10$. Thus, if x is the solution, then $9 = x^3 \pmod{11}$ implies

$$9^7 = (x^3)^7 = x^{3 \cdot 7} = x^{1+2 \cdot 10} = x \cdot (x^{10})^2 = x \cdot (1)^2 = x \pmod{11}.$$

Thus, $x = 9^7$. Now we must find $x \in \{0, 1, \dots, 10\}$ so that $x = 9^7$.

- $9^2 = 81 = 4 \pmod{11}$
- $9^4 = (9^2)^2 = 4^2 = 16 = 5 \pmod{11}$.
- $9^7 = 9^4 \cdot 9^2 \cdot 9 = 5 \cdot 4 \cdot 9 = 180 = 176 + 4 = 4 \pmod{11}$.

Thus, $x = 4$ is the solution. We can also check $4^3 = 64 = 55 + 9 = 9 \pmod{11}$.

12.4.1 Diophantine equations

As congruences relate to divisibility, we can also use them to solve diophantine equations.

Example 12.10 What are the integer solutions to $9x^2 + 9x + 2 = y^4$?

We will present two methods for comparison, first using the usual divisibility methods from last week, and then a second method using modular arithmetic.

Method 1: If x, y are integer solutions, then

$$y^4 = 9x^2 + 9x + 2 = (3x+2)(3x+1).$$

Since $y^4 \geq 0$, we know $3x+1 > 0$ and $3x+2 > 0$ or $3x+1 < 0$ and $3x+2 < 0$. First assume $3x+1 > 0$ and $3x+2 > 0$. We know that $3x+1$ and $3x+2$ are coprime since $\gcd(3x+1, 3x+2)$ divides $(3x+2) - (3x+1) = 1$. Thus, by Theorem 11.18, $3x+1 = a^4$ and $3x+2 = b^4$ for some integers a and b . (We can assume they are non-negative). But then $b^4 - a^4 = 3x+2 - (3x+1) = 1$. This is only possible if $(a, b) = (0, 1)$ (exercise). But we can't have $a = 0$ since then $3x+1 = 0$, which is impossible if x is an integer. The same is true in the case where $3x+1 < 0$ and $3x+2 < 0$. Thus, there are no solutions to the given equation.

As you can see, the above solution is quite long (and we didn't do some of the details). Now let's instead use modular arithmetic for a shorter proof:

Method 2: If x, y are integer solutions to $9x^2 + 9x + 2 = y^4$, then

$$y^4 = 9x^2 + 9x + 2 = 2 \pmod{3}.$$

Let $z \in \{0, 1, 2\}$ be such that $z = y \pmod{3}$, then $z^4 = y^4 = 2 \pmod{3}$. So let's test some z 's!

- $0^4 = 0$
- $1^4 = 1$
- $2^4 = 16 = 1 \pmod{3}$.

So there are no $z \in \{0, 1, 2\}$ so that $z^4 = 2 \pmod{3}$. Thus, there are no integer solutions to $9x^2 + 9x + 2 = y^4$.

12.5 Exercises

The relevant exercises in Liebeck's book are in Chapters 13 and 14.

■ **Exercise 12.3** Determine whether the following equations have integer solutions:

- $x^2 + y^2 = 9z + 3$.
- $3x^2 - y^2 = -2$.
- $3x^2 + 2 = y^2 + 6z^3$.

■ **Exercise 12.4** Show that If $n \in \mathbb{N}$, then n^5 and n have the same last digit (that is, the digit in the 1's place).

■ **Exercise 12.5** Find the last digit of $7^{7^{7^7}}$.

■ **Exercise 12.6** Suppose $n \in \mathbb{N}$ satisfies $n = 3 \pmod{4}$. Show that n has a prime factor p with $p = 3 \pmod{4}$.

■ **Exercise 12.7** Show that there are no integer solutions to $m^3 - n^3 = 3$ or $m^3 - n^3 = 4$. (Hint: if there were solutions, there would also be a solution $\pmod{7}$).

■ **Exercise 12.8** Show that for all primes $p > 3$, $p = \pm 1 \pmod{6}$. Use this to show 24 divides $p^2 - 1$ for all primes $p > 3$.

■ **Exercise 12.9** Show that if $p_1 < \dots < p_{31}$ are prime and 30 divides $p_1^4 + \dots + p_{31}^4$, then $p_1 = 2$, $p_2 = 3$ and $p_3 = 5$. *Hint: Assume the contrary and use FLT.*

■ **Exercise 12.10** (Regional Mathematical Olympiad 1998, India) Show that if $5 < p_1 < \dots < p_n$ and 6 divides $p_1^2 + \dots + p_n^2$, then 6 divides n .

■ **Exercise 12.11** Given that p and $8p^2 + 1$ are prime, find p .

■ **Exercise 12.12** Prove that 42 divides $n^7 - n$ for all $n \in \mathbb{N}$.

- **Exercise 12.13** If 7 does not divide a , then 7 divides $a^3 - 1$ or 7 divides $a^3 + 1$.